

Setup Guide - Space Shooter (Unity 2022.3 LTS)

1) Create/Open Project

- Open this repository in Unity Hub.
- Use Unity **2022.3 LTS**.

2) Create Main Scene

- Create `Assets/Scenes/Main.unity`.
- Add an orthographic camera and black background.

3) Add Managers

Create empty GameObject `Managers`, then child objects:

- `InputHandler` + `InputHandler.cs`
- `GameManager` + `GameManager.cs`
- `ScoreSystem` + `ScoreSystem.cs`
- `PowerUpSystem` + `PowerUpSystem.cs`
- `SoundManager` + `AudioSource` + `SoundManager.cs`
- `EnemySpawner` + `EnemySpawner.cs`

Assign references in `GameManager` :

- Enemy Spawner
- Player Controller
- Score System

4) Create Player

- `SpriteRenderer` + ship sprite
- `Rigidbody2D` (Kinematic)
- `Collider2D` (trigger = false)
- Scripts:
 - `PlayerController`
 - `PlayerHealth`
 - `CollisionHandler` (role = Player)
 - `FactionAffiliation` (Faction = Player)
- Child object `FirePoint` slightly above ship center.

Assign in `PlayerController` :

- Bullet prefab
- Fire point
- `PlayerHealth` reference

5) Create Enemy Prefab

- SpriteRenderer + enemy sprite
- Rigidbody2D (Kinematic)
- Collider2D
- Scripts:
 - EnemyAI
 - Damageable
 - CollisionHandler (role = Enemy)
 - FactionAffiliation (Faction = Enemy)
- Child FirePoint below enemy center.
- Save as prefab and assign to EnemySpawner.enemyPrefab.

6) Create Bullet Prefabs

Player bullet prefab:

- Sprite + trigger collider
- Bullet script (damage 10, speed 14)

Enemy bullet prefab:

- Sprite + trigger collider
- Bullet script (damage 8, speed 9)

Assign bullet prefabs in:

- PlayerController (player bullet)
- EnemyAI (enemy bullet)

7) Create Power-up Prefabs

Create 3 prefabs with sprite + trigger collider + PowerUpPickup :

- Shield
- RapidFire
- HealthRestore

Assign them to PowerUpSystem fields.

8) UI Setup

Create Canvas and panels:

- Main Menu (Start, Quit)
- HUD (Health text, Score text, Wave text, optional status icons)
- Pause panel (Resume, Menu)
- Game Over panel (Final score, Restart, Menu)

Attach UIManager and wire button events:

- Start -> UIManager.OnStartButtonPressed
- Resume -> UIManager.OnResumeButtonPressed
- Restart -> UIManager.OnRestartButtonPressed
- Main Menu -> UIManager.OnMainMenuButtonPressed
- Quit -> UIManager.OnQuitButtonPressed

9) Parallax Background

- Add two background layers.
- Add `ParallaxScroller` to any object.
- Assign near/far layer transforms.

10) Build Windows Executable

- File -> Build Settings
- Platform: PC, Mac & Linux Standalone
- Target: Windows x86_64
- Add `Main.unity` scene
- Build